

บรรณานุกรม

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ภาคผนวก

ภาคผนวก ก. รายละเอียดโปรแกรม

" Code ทั้งหมด ใน Form1.Vb ของ Interface Project Vb.net

```
Imports System.Data
Imports System.Data.SqlClient
Imports System.Globalization
Imports System.Drawing.Printing
Imports System.Text

Public Class Form1
    Inherits System.Windows.Forms.Form

    #Region " Windows Form Designer generated code "

    #End Region

    Dim Conn As SqlConnection = New SqlConnection
    Dim da As SqlDataAdapter
    Dim ds As DataSet = New DataSet
    Dim dt As DataTable

    Dim strConn As String = "server =(local);initial catalog=project;integrated
security=SSPI;persist security info=False; workstation id=THAIVB;packet
size=4096"

    Public oCP As New Rs232

    Public intCommPort, intBaud, intData As Integer
    Public bytStop As Rs232.DataStopBit
    Public bytParity As Rs232.DataParity

    Private SendData As String = ""
    Private Sendloop As Integer = 0
    Private TextSend As String = ""

    Private CardIn_out As Boolean = False
    Private writewd As String = "write" + Chr(13) + Chr(5)
    Private loopw As Integer = 0
    Private StatusWrite As Boolean = False

    Private Wait As WaitUpdate

    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Load

        Me.WindowState = FormWindowState.Maximized
        Cursor.Current = Cursors.WaitCursor
```

```

StatusWrite = False
Wait = New WaitUpdate

Me.tmrWrite.Enabled = False
With Conn
    If .State = ConnectionState.Open Then .Close()
    .ConnectionString = strConn
    .Open()
End With

Dim i As Integer
For i = 0 To 3
    mComs(i) = IsPortAvailable(i + 1)
Next
InitPort()
Dim Login As New LoginUser
If Login.ShowDialog Then
    If StatusUser Then
        StdID.ReadOnly = False
        UserState.Text = "      เจ้าหน้าที่"
        DataSubject.Enabled = True
    Else
        StdID.ReadOnly = True
        UserState.Text = "      นักศึกษา"
        DataSubject.Enabled = False
    End If
End If
If Student_ID = "" Then
    ResetButton()
End If
If StatusUser Then
    Regis.Visible = True
End If
tmrCardout.Enabled = True

End Sub
Private Function InitPort() As Boolean
    intCommPort = 1
    intBaud = 9600
    intData = 8
    bytParity = Rs232.DataParity.Parity_None
    bytStop = Rs232.DataStopBit.StopBit_1
    oCP.Open(intCommPort, intBaud, intData, bytParity, bytStop, 4096)

    tmrRead.Enabled = True

End Function
Private Function IsPortAvailable(ByVal ComPort As Integer) As Boolean
    Try
        oCP.Open(ComPort, 9600, 8, Rs232.DataParity.Parity_None, _
            Rs232.DataStopBit.StopBit_1, 4096)

        oCP.Close()
        Return True
    Catch
        Return False
    End Try

```

```

        End Try
    End Function

    Private Sub Button1_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Button1.Click
        Send()
    End Sub
    Private Function Send() As Boolean
        SendData = ReadText.Text
        Try
            ' เขียน string ออก port
            oCP.Write(Encoding.ASCII.GetBytes(Me.ReadText.Text & Chr(13)))
            WriteMessage(CStr(SendData), True)
        Catch ex As Exception
            MessageBox.Show("Unable to write to comm port")
        Finally
            ReadText.Text = ""
            ReadText.Focus()
        End Try
    End Function

    Private Sub MenuItem2_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem2.Click
        Me.Close()
    End Sub

    Private Sub MenuItem4_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem4.Click
        Dim Form As New Resume_STD
        Form.Show()
    End Sub

    Private Sub MenuItem5_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem5.Click
        Dim Form As New Resume_Fa1
        Form.Show()
    End Sub

    Private Sub MenuItem6_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem6.Click
        Dim Form As New Resume_F2
        Form.Show()
    End Sub

    Private Sub MenuItem9_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem9.Click
        Dim Form As New STD_STATUS
        Form.Show()
    End Sub

    Private Sub MenuItem11_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem11.Click
        Dim Form As New Change_SUB
        Form.Show()
    End Sub

```

```

Private Sub MenuItem12_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem12.Click
    Dim Form As New GRADE
    Form.Show()
End Sub

```

```

Private Sub MenuItem13_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem13.Click
    If StatusUser Then
        Dim Form As New Add_SUBJECT
        Form.Show()
    Else
        MessageBox.Show("คุณไม่ใช่เจ้าหน้าที่", "คำเตือน", MessageBoxButtons.OK,
MessageBoxIcon.Warning)
    End If
End Sub

```

```

Private Sub MenuItem14_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs)
    If StatusUser Then
        Dim Form As New Add_Teacher
        Form.Show()
    Else
        MessageBox.Show("คุณไม่ใช่เจ้าหน้าที่", "คำเตือน", MessageBoxButtons.OK,
MessageBoxIcon.Warning)
    End If
End Sub

```

```

Private Sub MenuItem16_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem16.Click
    If StatusUser Then
        Dim Form As New Add_RELIGION
        Form.Show()
    Else
        MessageBox.Show("คุณไม่ใช่เจ้าหน้าที่", "คำเตือน", MessageBoxButtons.OK,
MessageBoxIcon.Warning)
    End If
End Sub

```

```

Private Sub MenuItem17_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem17.Click
    If StatusUser Then
        Dim Form As New Add_FACILITY
        Form.Show()
    Else
        MessageBox.Show("คุณไม่ใช่เจ้าหน้าที่", "คำเตือน", MessageBoxButtons.OK,
MessageBoxIcon.Warning)
    End If
End Sub

```

```

Private Sub MenuItem18_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem18.Click
    If StatusUser Then
        Dim Form As New Add_DEPART
        Form.Show()
    Else

```

```

        MessageBox.Show("คุณไม่ใช่เจ้าหน้าที่", "คำเตือน", MessageBoxButtons.OK,
        MessageBoxIcon.Warning)
    End If
End Sub

Private Sub tmrRead_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles tmrRead.Tick
    Try
        While (oCP.Read(1) <> -1)
            Me.tmrRead.Enabled = False
            If (Chr(oCP.InputStream(0)) <> Chr(4)) Then
                WriteMessage(Chr(oCP.InputStream(0)), False)
            Else
                CardIn_out = True
            End If
            CardIn_out = True
        End While

        Catch exc As Exception
            If StatusWrite Then
                StatusWrite = False
                TextSend = TxtSendData
                Me.tmrWrite.Enabled = True
                Sendloop = 0

            Else
                If CardIn_out And ReadText.Text.Length > 100 Then
                    CheckCardIn()
                    CardIn_out = False
                Else
                    Me.tmrRead.Enabled = True
                End If
            End If

        End Try

    End Sub
Private Function ChRegisCard() As Boolean
    Dim strlen As Integer = ReadText.Text.Length
    Dim loopstr As Integer = 0
    While (loopstr < strlen)
        If ReadText.Text.Chars(loopstr) = Chr(3) Then
            Return True
        End If
        loopstr = loopstr + 1
    End While
    Return False
End Function
Private Function CardIDin() As Boolean
    Dim temp As String = ReadText.Text
    Dim i As Integer = 0
    While i < temp.Length
        If temp.Chars(i) <> Chr(2) Then
            UniqueCardID += temp.Chars(i)
        Else
            Exit Function
        End If
    End While
End Function

```

```

        i += 1
    End While
End Function
Private Function CheckCardIn() As Boolean
    'CardIDin()
    If ChRegisCard() Then

        Dim CardIn As String = ReadText.Text
        SmartHealth = ""
        SmartStudent = ""
        SmartMoney = ""

        CardIn = CardIn + Chr(4)
        Dim LoopIn As Integer = 0
        While CardIn.Chars(LoopIn) <> Chr(4)
            If CardIn.Chars(LoopIn) = Chr(2) Then
                LoopIn = LoopIn + 1
                While CardIn.Chars(LoopIn) <> Chr(3)
                    SmartStudent = SmartStudent + CardIn.Chars(LoopIn)
                    LoopIn = LoopIn + 1
                End While
                LoopIn = LoopIn + 1
                While CardIn.Chars(LoopIn) <> Chr(3)
                    SmartHealth = SmartHealth + CardIn.Chars(LoopIn)
                    LoopIn = LoopIn + 1
                End While
                LoopIn = LoopIn + 1
                While CardIn.Chars(LoopIn) <> Chr(3)
                    SmartMoney = SmartMoney + CardIn.Chars(LoopIn)
                    LoopIn = LoopIn + 1
                End While
                LoopIn = LoopIn + 1
            End If
            LoopIn = LoopIn + 1
        End While

        CheckPassword()
        Me.tmrCardout.Enabled = True
    Else
        MsgBox("บัตร ยังไม่ได้ลงทะเบียน")
        Me.tmrCardout.Enabled = True
    End If

    Return True
End Function
Private Function CheckPassword() As Boolean

    Dim loopdata As Integer = 0

    Dim datapwd As String = ""
    Dim dataIdstd As String = ""
    Dim tempname As String = ""
    Dim tempsurname As String = ""

    Dim lengthdata As Integer = SmartStudent.Length

    While (loopdata < lengthdata)

```



```

While (SmartStudent.Chars(loopdata) <> Chr(64))
    datapwd = datapwd + SmartStudent.Chars(loopdata)
    loopdata = loopdata + 1
End While
loopdata = loopdata + 1

While (SmartStudent.Chars(loopdata) <> Chr(64))
    dataIdstd = dataIdstd + SmartStudent.Chars(loopdata)
    loopdata = loopdata + 1
End While
loopdata = loopdata + 1

While (SmartStudent.Chars(loopdata) <> Chr(64))
    tempname = tempname + SmartStudent.Chars(loopdata)
    loopdata = loopdata + 1
End While
loopdata = loopdata + 1

While (loopdata < lengthdata)
    tempsurname = tempsurname + SmartStudent.Chars(loopdata)
    loopdata = loopdata + 1
End While
loopdata = loopdata + 1
End While

Student_ID = dataIdstd
thaiNM = tempname
ThaiSurNM = tempsurname
Dim temp As String = DecodePwd(datapwd)
CheckPassword(temp)

End Function

Private Function SubString(ByVal data As String) As String
    Dim i As Integer = 0

    Dim Strtemp As String = ""
    While data.Chars(i) <> Chr(2)
        Strtemp = Strtemp + data.Chars(i)
        i = i + 1
    End While
    i = i + 1
    Student_ID = ""
    While data.Chars(i) <> Chr(3)
        Student_ID = Student_ID + data.Chars(i)
        i = i + 1
    End While
    Return Strtemp
End Function

Public Function CheckPassword(ByVal Data As String) As Boolean
    DataOfCard = Data
    Dim Forms As New LOGIN
    If Forms.ShowDialog() Then
        If Pwd Then
            ShowStatus()
        End If
    End If
End Function

```

```

End Function
Private Function ShowStatus() As Boolean
    Dim sql As String = ""
    sql = "SELECT a.Std_ID, a.Std_BLOOD, b.NAME as Name1, c.NAME as Name2 "
    sql = sql + " FROM Student a, Department b, Factory c, Status_ST d"
    sql = sql + " where a.Sta_NUM = d.Sta_NUM and d.Sta_DIV = b.Sta_DIV and
b.Sta_Fact = c.Sta_Fact "
    sql = sql + " and a.Std_ID = '" + Student_ID + "'"

    da = New SqlDataAdapter(sql, Conn)
    ds.Tables.Clear()
    da.Fill(ds, "Query1")
    If ds.Tables("Query1").Rows.Count <> 0 Then
        StdID.Text = CStr(ds.Tables("Query1").Rows(0).Item("Std_ID"))
        Factory.Text = CStr(ds.Tables("Query1").Rows(0).Item("Name2"))
        Depart.Text = CStr(ds.Tables("Query1").Rows(0).Item("Name1"))
        Blood.Text = CStr(ds.Tables("Query1").Rows(0).Item("Std_BLOOD"))
        GlDepart = Depart.Text
        GlFactory = Factory.Text

        sql = "Select c.Std_NAME, c.Std_SURNAME, c.StdE_NAME, c.StdE_SURNAME"
        sql = sql + " From Student a, NameStudent c "
        sql = sql + " Where a.Std_NUM = c.Std_NUM "
        sql = sql + " and a.Std_ID = '" + Student_ID + "'"

        da.SelectCommand.CommandText = sql
        da.Fill(ds, "Query2")
        If ds.Tables("Query2").Rows.Count <> 0 Then
            ThaiName.Text =
CStr(ds.Tables("Query2").Rows(0).Item("Std_NAME"))
            ThaiSurname.Text =
CStr(ds.Tables("Query2").Rows(0).Item("Std_SURNAME"))
            thaiNM = ThaiName.Text
            ThaiSurNM = ThaiSurname.Text
            EThaiSurNM =
CStr(ds.Tables("Query2").Rows(0).Item("StdE_SURNAME"))
            EthaiNM = CStr(ds.Tables("Query2").Rows(0).Item("StdE_NAME"))
            ShowButton()
        End If
    Else
        MessageBox.Show("No Record !", "ถ้าเตือน", MessageBoxButtons.OK,
MessageBoxIcon.Error)
    End If
End Function
Private Sub WriteMessage(ByVal message As String, ByVal write As Boolean)
    If write Then
    Else
        ReadText.Text += message ' String ทั้งหมดที่อ่านเข้ามา อยู่ในตัวแปร inputString
    End If
End Sub

Private Sub Bexit_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Bexit.Click
    Me.Close()
End Sub

```

```

Private Sub MenuItem19_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem19.Click
    Dim Card As New RegisCard
    If StatusUser Then
        If Student_ID <> "" Then
            If Card.ShowDialog Then
                If PwdToCard <> "" Then
                    EncodePwd()
                End If
            End If
        Else
            Dim Student As New Resume_STD
            Student.Show()
        End If
    Else
        MessageBox.Show("คุณไม่ใช่เจ้าหน้าที่", "คำเตือน", MessageBoxButtons.OK,
MessageBoxIcon.Warning)
    End If
End Sub
Private Function EncodePwd() As Boolean
    If SmartHealth = "" Then
        SmartHealth = "none" 'ว่าง
    End If
    If SmartMoney = "" Then
        SmartMoney = "0" 'ว่าง
    End If
    Dim temp As String = EncoderPwd(PwdToCard)

    Wait.Show()
    TxtSendData = temp + Chr(64) + Student_ID + Chr(64) + EthaiNM + Chr(64)
+ EThaiSurNM + Chr(3) + SmartHealth + Chr(3) + SmartMoney + Chr(3) + Chr(4) +
Chr(13) + Chr(5)
    TextSend = "write" + Chr(13) + Chr(5)
    StatusWrite = True
    tmrCardout.Enabled = False
    Me.tmrWrite.Enabled = True
    Sendloop = 0

End Function
Private Function EncoderPwd(ByVal data As String) As String

    Dim i As Integer = 4
    Dim temp0 As Integer = Asc(data.Chars(0))
    Dim temp1 As Integer = Asc(data.Chars(1))
    Dim temp2 As Integer = Asc(data.Chars(2))
    Dim temp3 As Integer = Asc(data.Chars(3))
    Dim CodeData As String = ""
    CodeData = Chr(temp0 - 10)
    CodeData = CodeData + Chr(temp1 - 15)
    CodeData = CodeData + Chr(temp2 - 10)
    CodeData = CodeData + Chr(temp3 - 17)
    While i < data.Length
        CodeData = CodeData + data.Chars(i)
        i = i + 1
    End While
    Return CodeData

```

```
End Function
Private Function DecodePwd(ByVal CodeData As String) As String
```

```
    Dim i As Integer = 0
    Dim tenp0 As Integer = Asc(CodeData.Chars(0))
    Dim tenp1 As Integer = Asc(CodeData.Chars(1))
    Dim tenp2 As Integer = Asc(CodeData.Chars(2))
    Dim tenp3 As Integer = Asc(CodeData.Chars(3))
    Dim CodeData1 As String = ""
    i = 4

    CodeData1 = Chr(tenp0 + 10)
    CodeData1 = CodeData1 + Chr(tenp1 + 15)
    CodeData1 = CodeData1 + Chr(tenp2 + 10)
    CodeData1 = CodeData1 + Chr(tenp3 + 17)
    While i < CodeData.Length
        CodeData1 = CodeData1 + CodeData.Chars(i)
        i = i + 1
    End While
    Return CodeData1
```

```
End Function
```

```
Private Sub StdID_TextChanged(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles StdID.TextChanged
    If StdID.TextLength = 8 Then
        Student_ID = StdID.Text
        Dim sql As String = ""
        sql = "Select * from Student Where Std_ID = '" + Student_ID + "'"
        da = New SqlDataAdapter(sql, Conn)
        ds.Tables.Clear()
        da.Fill(ds, "Query1")
        If ds.Tables("Query1").Rows.Count <> 0 Then
            ShowStatus()
        Else
            ClearText()
            If MessageBox.Show("รหัสนักศึกษาไม่มีในฐานข้อมูล! คุณต้องการที่จะทำการเพิ่มข้อมูลรหัสนักศึกษาใหม่หรือไม่",
"คำถาม", MessageBoxButtons.YesNo, MessageBoxIcon.Stop) = DialogResult.Yes Then

                AddNewStudent()

            End If
        End If
    Else
        ClearText()
        Student_ID = ""
    End If
End Sub
Private Function AddNewStudent() As Boolean
    RegisterSTD = False
    Dim Std As New Resume_STD
    If Std.ShowDialog() Then

        If RegisterSTD Then
            RegisterSTD = False
```

```

Dim fa As New Resume_Fa
If fa.ShowDialog() Then

    If RegisterSTD Then
        RegisterSTD = False
        Dim Ma As New Resume_Ma
        If Ma.ShowDialog() Then

            If RegisterSTD Then
                RegisterSTD = False
                Dim Fa2 As New Resume_Fa1
                If Fa2.ShowDialog() Then

                    If RegisterSTD Then
                        RegisterSTD = False
                        Dim StdState As New STD_STATUS
                        If StdState.ShowDialog() Then

                            If RegisterSTD Then
                                RegisterSTD = False
                                ShowStatus()
                                Return True
                            End If
                        End If
                    End If
                End If
            End If
        End If
    End If
End If

DelCancel()
Return False

End Function
Private Function DelCancel() As Boolean
    Dim sql As String = ""
    Dim Stanum As String = ""
    Dim RstdNum As String = ""
    Dim StdNum As String = ""

    sql = "Select Std_ID,Sta_NUM,Adr_NUMStd,Std_NUM from Student Where
Std_ID = '" + StdID.Text + "'"
    da = New SqlDataAdapter(sql, Conn)
    ds.Tables.Clear()
    da.Fill(ds, "Query1")
    If ds.Tables("Query1").Rows.Count <> 0 Then
        Stanum = ds.Tables("Query1").Rows(0).Item("Sta_NUM")
        RstdNum = ds.Tables("Query1").Rows(0).Item("Adr_NUMStd")
        StdNum = ds.Tables("Query1").Rows(0).Item("Std_NUM")

        sql = "Delete From Student where Std_ID = '" + StdID.Text + "'"
        da.SelectCommand.CommandText = sql
        da.Fill(ds, "delete")
    End If
End Function

```

```

    sql = "Delete From NameStudent where Std_NUM = " + StdNum
    da.SelectCommand.CommandText = sql
    da.Fill(ds, "delete")

    sql = "Delete From AddrStudent where Adr_NUMStd = " + RstdNum
    da.SelectCommand.CommandText = sql
    da.Fill(ds, "delete")

End If

sql = "SELECT a.Std_ID,a.Fa_NUM,b.Std_NUM FROM FatherID a INNER JOIN "
sql = sql + " Father b ON a.Fa_NUM = b.Fa_NUM WHERE (a.Std_ID = '" +
StdID.Text + "') "
da = New SqlDataAdapter(sql, Conn)
ds.Tables.Clear()
da.Fill(ds, "Query1")
If ds.Tables("Query1").Rows.Count = 3 Then

    Dim temp As String = ds.Tables("Query1").Rows(0).Item("Std_NUM")
    sql = "Delete From NameStudent where Std_NUM = " + temp
    da.SelectCommand.CommandText = sql
    da.Fill(ds, "delete")

    temp = ds.Tables("Query1").Rows(1).Item("Std_NUM")
    sql = "Delete From NameStudent where Std_NUM = " + temp
    da.SelectCommand.CommandText = sql
    da.Fill(ds, "delete")

    temp = ds.Tables("Query1").Rows(2).Item("Std_NUM")
    sql = "Delete From NameStudent where Std_NUM = " + temp
    da.SelectCommand.CommandText = sql
    da.Fill(ds, "delete")

    temp = ds.Tables("Query1").Rows(0).Item("Fa_NUM")
    sql = "Delete From father where fa_NUM = " + temp
    da.SelectCommand.CommandText = sql
    da.Fill(ds, "delete")

    temp = ds.Tables("Query1").Rows(1).Item("Fa_NUM")
    sql = "Delete From father where fa_NUM = " + temp
    da.SelectCommand.CommandText = sql
    da.Fill(ds, "delete")

    temp = ds.Tables("Query1").Rows(2).Item("Fa_NUM")
    sql = "Delete From father where fa_NUM = " + temp
    da.SelectCommand.CommandText = sql
    da.Fill(ds, "delete")

    sql = "Delete From fatherID where Std_ID = '" + StdID.Text + "'"
    da.SelectCommand.CommandText = sql
    da.Fill(ds, "delete")

End If

If Stanum <> "" Then
    sql = "Select * from Status_ST Where Sta_NUM = " + Stanum
    da = New SqlDataAdapter(sql, Conn)

```

```

        ds.Tables.Clear()
        da.Fill(ds, "Query1")
        If ds.Tables("Query1").Rows.Count <> 0 Then

            sql = "Delete From Status_ST where Sta_NUM = " + Stanum
            da.SelectCommand.CommandText = sql
            da.Fill(ds, "delete")

        End If
    End If
    StdID.Text = ""
End Function
Private Function ClearText() As Boolean
    ThaiName.Clear()
    ThaiSurname.Clear()
    Factory.Clear()
    Depart.Clear()
    Blood.Clear()
    ResetButton()
    Return True
End Function
Private Function ResetButton() As Boolean
    DataStd.Enabled = False
    'DataSubject.Enabled = False
    DataHealth.Enabled = False
    EditSubject.Enabled = False
    CheckGrade.Enabled = False
    Bfather.Enabled = False
    BfaTher2.Enabled = False
    BStdState.Enabled = False
End Function
Private Function ShowButton() As Boolean
    DataStd.Enabled = True
    'DataSubject.Enabled = True
    DataHealth.Enabled = True
    EditSubject.Enabled = True
    CheckGrade.Enabled = True
    Bfather.Enabled = True
    BfaTher2.Enabled = True
    BStdState.Enabled = True
End Function

Private Sub CheckGrade_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles CheckGrade.Click
    Dim grade As New GRADE
    grade.Show()
End Sub

Private Sub EditSubject_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles EditSubject.Click
    Dim EditRegis As New Change_SUB
    EditRegis.Show()
End Sub

Private Sub DataStd_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles DataStd.Click

```

```

        Dim Std As New Resume_STD
        Std.Show()
    End Sub

    Private Sub DataSubject_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles DataSubject.Click
        Dim Subject As New Add_SUBJECT
        Subject.Show()
    End Sub

    Private Sub MenuItem20_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem20.Click
        Dim Subject As New Add_Status_SUB
        Subject.Show()
    End Sub

    Private Sub Bfather_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Bfather.Click
        Dim far As New Resume_F2
        If far.ShowDialog Then

            End If
    End Sub

    Private Sub BfaTher2_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles BfaTher2.Click
        Dim far As New Resume_Fa1
        far.ShowDialog()

    End Sub

    Private Sub BStdState_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles BStdState.Click
        Dim Sta As New STD_STATUS
        Sta.ShowDialog()

    End Sub

    Private Sub Regis_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Regis.Click
        StdID.Clear()
        NewID = True
        AddNewStudent()
        NewID = False
    End Sub

    Private Sub tmrWrite_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles tmrWrite.Tick

        If TextSend.Chars(Sendloop) <> Chr(5) Then
            ReadText.Text = TextSend.Chars(Sendloop)
            Sendloop = Sendloop + 1

            Try
                oCP.Write(Encoding.ASCII.GetBytes(Me.ReadText.Text))
                WriteMessage(ReadText.Text, True)
            Catch ex As Exception

```



```

        MessageBox.Show("Unable to write to comm port")
        Me.tmrWrite.Enabled = False
    Finally
        ReadText.Text = ""
    End Try
Else
    Me.tmrWrite.Enabled = False
    Sendloop = 0
    If StatusWrite Then
        Me.tmrRead.Enabled = True
    Else
        'MsgBox("Register Successfull.....")
        Wait.Close()
        Wait = New WaitUpdate
        Me.tmrCardout.Enabled = True
    End If
End If

End Sub

Private Sub tmrCardout_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles tmrCardout.Tick
    Dim out As Boolean = False
    Try
        While (oCP.Read(1) <> -1)
            Me.tmrCardout.Enabled = False
            WriteMessage(Chr(oCP.InputStream(0)), False)
            out = True
        End While
    Catch exc As Exception
        If out Then
            If ReadText.Text.Length > 15 Then
                If ReadText.Text.Chars(1) = Chr(82) Then
                    out = False
                    CardIn_out = False
                    tmrWrite.Enabled = False
                    Me.tmrCardout.Enabled = False
                    Me.tmrCardout.Stop()
                    'MsgBox("Remove Card out")
                    StdID.Text = ""
                    ReadText.Text = ""
                    ResetButton()
                    tmrRead.Enabled = True
                Else
                    out = False
                    Me.tmrCardout.Enabled = True
                End If
            Else
                out = False
                Me.tmrCardout.Enabled = True
            End If
        Else
            Me.tmrCardout.Enabled = True
            ReadText.Text = Chr(13)
        End If
    End Try
End Sub

```

```

        Try
            oCP.Write(Encoding.ASCII.GetBytes(Me.ReadText.Text &
Chr(13)))
            WriteMessage(ReadText.Text, True)
        Catch ex As Exception
            MessageBox.Show("Unable to write to comm port")
            Me.tmrWrite.Enabled = False
        Finally
            'ReadText.Text = ""
        End Try
    End If
End Try

```

```
End Sub
```

```
End Class
```

```
.....
```

Code ทั้งหมด ใน Form1.Vb ของ SmartMoneyCard Project Vb.net

```

Imports System.Data
Imports System.Data.SqlClient
Imports System.Globalization
Imports System.Drawing.Printing
Imports System.Text
Public Class Form1
    Inherits System.Windows.Forms.Form

    #Region " Windows Form Designer generated code "

    #End Region

    Dim Conn As SqlConnection = New SqlConnection
    Dim da As SqlDataAdapter
    Dim ds As DataSet = New DataSet
    Dim dt As DataTable

    Public oCP As New Rs232
    Public intCommPort, intBaud, intData As Integer
    Public bytStop As Rs232.DataStopBit
    Public bytParity As Rs232.DataParity
    Public mComs(3) As Boolean ' ตรวจสอบ Serial Port

    Dim CardIn_out As Boolean
    Dim Sendloop As Integer = 0
    Dim TextSend As String = ""
    Dim StatusWrite As Boolean
    Dim Wait As WaitUpdate

    Dim InfoDate As DateTimeFormatInfo

    ' print document*****

```

```

Dim UseFont As New Font("MS Sans Serif", 10)
Dim prDlg As PrintDialog = New PrintDialog
Private prDoc As PrintDocument = New PrintDocument

'*/*****

Private Sub BnExit_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles BnExit.Click
    Me.Close()
End Sub

Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Load
    Me.WindowState = FormWindowState.Maximized
    Cursor.Current = Cursors.WaitCursor
    tmrDate.Enabled = True
    Wait = New WaitUpdate
    TextMoney.Enabled = False
    BnSearch.Enabled = False
    BnSave.Enabled = False
    InfoDate = DateTimeFormatInfo.InvariantInfo

    With Conn
        If .State = ConnectionState.Open Then .Close()
        .ConnectionString = strConn
        .Open()
    End With

    Dim i As Integer
    For i = 0 To 3
        mComs(i) = IsPortAvailable(i + 1)
    Next
    InitPort()

    AddHandler prDoc.PrintPage, New PrintPageEventHandler(AddressOf
Me.StringToPrint_Print)
    prDoc.DocumentName = "เติมเงิน"
    prDlg.Document = prDoc
    Cursor.Current = Cursors.Default
    tmrCardOut.Enabled = True
    StateLogin.Text = "ทั่วไป"
    StatusUser = False

End Sub
Private Sub StringToPrint_Print(ByVal sender As Object, ByVal e As
PrintPageEventArgs)

    AnyString(e.Graphics, "รหัสนักศึกษา : " + Student_id, 100, 100)
    AnyString(e.Graphics, "ชื่อ : " + NameStudent + " " + SurName,
100, 120)
    AnyString(e.Graphics, "วันที่ : " + DateCur.Text, 100, 140)

```

```

        AnyString(e.Graphics,
".....", 100, 150)

        AnyString(e.Graphics, "-----
-----", 100, 160)
        AnyString(e.Graphics, "          รายการ", 100, 175)
        AnyString(e.Graphics, "จำนวนเงิน", 295, 175)
        AnyString(e.Graphics, "-----
-----", 100, 185)

        AnyString(e.Graphics, "เพิ่มจำนวนเงิน", 100, 210)
        AnyString(e.Graphics, TextMoney.Text, 295, 210)

        Dim CurrentYPosition As Integer = 230
        Dim MainMoneyTol As Double
        Dim temp As Double = MainMoney.Text
        temp += TextMoney.Text

        CurrentYPosition = CurrentYPosition + 40
        AnyString(e.Graphics, "-----
-----", 100, CurrentYPosition)
        CurrentYPosition = CurrentYPosition + 10
        AnyString(e.Graphics, "          จำนวนเงินคงเดิม", 100, CurrentYPosition)
        AnyString(e.Graphics, MainMoney.Text, 300, CurrentYPosition)
        CurrentYPosition = CurrentYPosition + 10
        AnyString(e.Graphics, "-----
-----", 100, CurrentYPosition)
        CurrentYPosition = CurrentYPosition + 10
        AnyString(e.Graphics, "          จำนวนเงินคงเหลือ", 100, CurrentYPosition)
        AnyString(e.Graphics, CStr(temp.ToString("##0.00")), 300,
CurrentYPosition)
        CurrentYPosition = CurrentYPosition + 10
        AnyString(e.Graphics, "-----
-----", 100, CurrentYPosition)

    End Sub
    Private Sub AnyString(ByVal g As Graphics, ByVal printString As
String, ByVal xPos As Integer, ByVal yPos As Integer)
        Dim anyPoint As New PointF(xPos, yPos)
        g.DrawString(printString, UseFont, Brushes.Black, anyPoint)
    End Sub
    Private Function IsInstallPrinter() As Boolean
        IsInstallPrinter = False
        If prDoc.PrinterSettings.PrinterName = "<no default printer>"
Then
            IsInstallPrinter = False
        Else
            IsInstallPrinter = True
        End If
    End Function

```

```

Private Function InitPort() As Boolean
    intCommPort = 1
    intBaud = 9600
    intData = 8
    bytParity = Rs232.DataParity.Parity_None
    bytStop = Rs232.DataStopBit.StopBit_1
    oCP.Open(intCommPort, intBaud, intData, bytParity, bytStop, 4096)

    tmrRead.Enabled = True

End Function
Private Function IsPortAvailable(ByVal ComPort As Integer) As Boolean
    Try
        oCP.Open(ComPort, 9600, 8, Rs232.DataParity.Parity_None, _
            Rs232.DataStopBit.StopBit_1, 4096)
        oCP.Close()
        Return True
    Catch
        Return False
    End Try
End Function

Private Sub tmrDate_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles tmrDate.Tick
    Dim DateCur1 As Date
    DateCur.Text = ""
    DateCur.Text = CStr(DateCur1.Today.ToLongDateString) + "      "
    DateCur.Text = DateCur.Text +
CStr(DateCur1.Today.Now.ToLongTimeString)
End Sub

Private Sub tmrRead_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles tmrRead.Tick
    Try
        While (oCP.Read(1) <> -1)
            Me.tmrRead.Enabled = False
            If (Chr(oCP.InputStream(0)) <> Chr(4)) Then
                ReadText.Text = ReadText.Text +
Chr(oCP.InputStream(0))
            End If
            CardIn_out = True
        End While

    Catch exc As Exception
        If StatusWrite Then
            StatusWrite = False
            TextSend = SmartStudent + Chr(3) + SmartHealth + Chr(3) +
CStr(InsertMoneyStatus) + Chr(3) + Chr(4) + Chr(13) + Chr(5)
            Me.tmrWrite.Enabled = True
            Sendloop = 0
        Else
            If CardIn_out And ReadText.Text.Length > 100 Then

```

```

        CheckCardIn()
        CardIn_out = False
    Else
        Me.tmrRead.Enabled = True
    End If
End If

End Try
End Sub
Private Function ChRegisCard() As Boolean
    Dim strlen As Integer = ReadText.TextLength
    Dim loopstr As Integer = 0
    While (loopstr < strlen)
        If ReadText.Text.Chars(loopstr) = Chr(3) Then
            Return True
        End If
        loopstr = loopstr + 1
    End While
    Return False
End Function
Private Function CheckCardIn() As Boolean
    If ChRegisCard() Then

        Dim CardIn As String = ReadText.Text
        SmartHealth = ""
        SmartStudent = ""
        SmartMoney = ""

        CardIn = CardIn + Chr(4)
        Dim LoopIn As Integer = 0
        While CardIn.Chars(LoopIn) <> Chr(4)
            If CardIn.Chars(LoopIn) = Chr(2) Then
                LoopIn = LoopIn + 1
                While CardIn.Chars(LoopIn) <> Chr(3)
                    SmartStudent = SmartStudent +
CardIn.Chars(LoopIn)
                    LoopIn = LoopIn + 1
                End While
                LoopIn = LoopIn + 1
            While CardIn.Chars(LoopIn) <> Chr(3)
                SmartHealth = SmartHealth + CardIn.Chars(LoopIn)
                LoopIn = LoopIn + 1
            End While
                LoopIn = LoopIn + 1
            While CardIn.Chars(LoopIn) <> Chr(3)
                SmartMoney = SmartMoney + CardIn.Chars(LoopIn)
                LoopIn = LoopIn + 1
            End While
                LoopIn = LoopIn + 1
            End If
            LoopIn = LoopIn + 1
        End While
    End While

```

```

        CheckPassword()
    Else
        MsgBox("บัตร ยังไม่มีเงิน")
    End If

    Return True
End Function
Private Function CheckPassword() As Boolean
    Dim lengthdata As Integer = SmartStudent.Length
    Dim loopdata As Integer = 0

    Dim datapwd As String = ""
    Dim dataIdstd As String = ""
    Dim tempname As String = ""
    Dim tempsurname As String = ""

    While (loopdata < lengthdata)
        While (SmartStudent.Chars(loopdata) <> Chr(64))
            datapwd = datapwd + SmartStudent.Chars(loopdata)
            loopdata = loopdata + 1
        End While
        loopdata = loopdata + 1

        While (SmartStudent.Chars(loopdata) <> Chr(64))
            dataIdstd = dataIdstd + SmartStudent.Chars(loopdata)
            loopdata = loopdata + 1
        End While
        loopdata = loopdata + 1

        While (SmartStudent.Chars(loopdata) <> Chr(64))
            tempname = tempname + SmartStudent.Chars(loopdata)
            loopdata = loopdata + 1
        End While
        loopdata = loopdata + 1

        While (loopdata < lengthdata)
            tempsurname = tempsurname + SmartStudent.Chars(loopdata)
            loopdata = loopdata + 1
        End While
        loopdata = loopdata + 1
    End While

    Student_id = dataIdstd
    NameStudent = tempname
    SurName = tempsurname
    StdID.Text = Student_id
    StdName.Text = NameStudent + " " + SurName
    MainMoney.Text = CInt(SmartMoney).ToString("##0.00")
    TextMoney.Enabled = True
    tmrCardOut.Enabled = True

```

```

End Function

Private Sub tmrCardOut_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles tmrCardOut.Tick
    Dim out As Boolean = False
    Try
        While (oCP.Read(1) <> -1)
            Me.tmrCardOut.Enabled = False
            ReadText.Text = ReadText.Text + Chr(oCP.InputStream(0))
            out = True
        End While

    Catch exc As Exception
        If out Then
            If ReadText.Text.Length > 15 Then
                If ReadText.Text.Chars(1) = Chr(82) Then
                    SmartStudent = ""
                    SmartHealth = ""
                    SmartMoney = ""
                    out = False
                    CardIn_out = False

                    StdID.Text = ""
                    StdName.Text = ""
                    TextMoney.Text = "0.00"
                    MainMoney.Text = "0.00"
                    BnSave.Enabled = False
                    TextMoney.Enabled = False
                    Me.tmrCardOut.Enabled = False
                    Me.tmrCardOut.Stop()
                    tmrRead.Enabled = True
                Else
                    out = False
                    Me.tmrCardOut.Enabled = True
                End If
            Else
                out = False
                Me.tmrCardOut.Enabled = True
            End If
        End If

    Else
        Me.tmrCardOut.Enabled = True
        ReadText.Text = Chr(13)
        Try
            oCP.Write(Encoding.ASCII.GetBytes(Me.ReadText.Text &
Chr(13)))

        Catch ex As Exception
            MessageBox.Show("Unable to write to comm port")
        Finally
            End Try
        End If
    End If

```



```

    End Try
End Sub

Private Sub tmrWrite_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles tmrWrite.Tick
    If TextSend.Chars(Sendloop) <> Chr(5) Then
        ReadText.Text = TextSend.Chars(Sendloop)
        Sendloop = Sendloop + 1

        Try
            oCP.Write(Encoding.ASCII.GetBytes(Me.ReadText.Text))
        Catch ex As Exception
            MessageBox.Show("Unable to write to comm port")
            Me.tmrWrite.Enabled = False
        Finally
            ReadText.Text = ""
        End Try
    Else
        Me.tmrWrite.Enabled = False
        Sendloop = 0
        If StatusWrite Then
            Me.tmrRead.Enabled = True
        Else
            Wait.Close()
            Wait = New WaitUpdate
            MainMoney.Text = InsertMoneyStatus
            TextMoney.Text = "0.00"
            Me.tmrCardOut.Enabled = True
        End If
    End If

End If
End Sub

Private Sub BnSave_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles BnSave.Click

    If MessageBox.Show("ต้องการเพิ่มจำนวนเงินในบัตร ใช้ หรือ ไม่ ", "คำยืนยัน",
MessageBoxButtons.YesNo, MessageBoxIcon.Question) = DialogResult.Yes Then
        Dim Sql As String = ""

        Dim DateCur As Date
        If IsInstallPrinter() = True Then
            prDoc.Print()
        Else
            MessageBox.Show("กรุณาติดตั้งเครื่อง Printer ก่อนสั่งพิมพ์เอกสาร", "ผลการ
ตรวจสอบ", MessageBoxButtons.OK, MessageBoxIcon.Information)
        End If
        Try
            Dim tempMon1 As Double = MainMoney.Text
            Dim tempMon2 As Double = TextMoney.Text
            InsertMoneyStatus = tempMon1 + tempMon2
        End Try
    End If
End Sub

```

```

        ' ลงฐานข้อมูล
        Sql = "insert into
CardMoney(Std_ID,Name,SurName,Money,DateCurrent) values('"
        Sql = Sql & Student_id & "',' & NameStudent & "',''"
& SurName & "',' & tempMon2 & "',' & DateCur.Today().ToString("d",
InfoDate) & "'"")"

        da = New SqlDataAdapter(Sql, Conn)
        ds.Tables.Clear()
        da.Fill(ds, "Insert")

        ' *****
        Try

            TextSend = "write" + Chr(13) + Chr(5)
            StatusWrite = True
            Wait.Show()
            tmrCardOut.Enabled = False
            tmrWrite.Enabled = True

        Catch ex As Exception

        End Try

        Catch ex As Exception
            MsgBox("ข้อมูลผิดพลาด.")
        End Try

    End If

End Sub

Private Sub TextMoney_TextChanged(ByVal sender As System.Object,
ByVal e As System.EventArgs) Handles TextMoney.TextChanged
    If StatusUser Then
        If TextMoney.Text <> "" Then
            Dim TextLen As Integer = TextMoney.Text.Length
            If TextMoney.Text.Chars(TextLen - 1) >= "0" And
TextMoney.Text.Chars(TextLen - 1) <= "9" Or TextMoney.Text.Chars(TextLen
- 1) = "." Then
                BnSave.Enabled = True
            Else
                MessageBox.Show("กรุณาใส่ ข้อมูล ให้ถูกต้อง", "คำเตือน",
MessageBoxButtons.OK, MessageBoxIcon.Warning)
                TextMoney.Text = ""
                BnSave.Enabled = False
            End If
        Else
            BnSave.Enabled = False
        End If
    End If

```

```

        End If

    End Sub

    Private Sub Login_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Login.Click

        If Login.Text = "Login" Then
            Dim form As New LOGIN
            If form.ShowDialog Then
                If StatusUser Then
                    StateLogin.Text = "เจ้าหน้าที่"
                    BnSearch.Enabled = True
                    Login.Text = "Logout"
                    If TextMoney.Text <> 0 Then
                        BnSave.Enabled = True
                    End If
                End If
            End If
        End If
    Else
        Login.Text = "Login"
        StateLogin.Text = "ทั่วไป"
        StatusUser = False
        BnSearch.Enabled = False
        BnSave.Enabled = False
    End If

End Sub

Private Sub BnSearch_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles BnSearch.Click
    Dim Form As New Search
    If Form.ShowDialog Then

    End If
End Sub
End Class

```

```

" Code ทั้งหมด ใน Form1.Vb ของ SmartFood Project Vb.net
Imports System.Data
Imports System.Data.SqlClient
Imports System.Globalization
Imports System.Drawing.Printing
Imports System.Text

Public Class Form1
    Inherits System.Windows.Forms.Form

```

```
#Region " Windows Form Designer generated code "
```

```
#End Region
```

```
Dim Conn As SqlConnection = New SqlConnection
Dim da As SqlDataAdapter
Dim ds As DataSet = New DataSet
Dim dt As DataTable
```

```
Public oCP As New Rs232
Public intCommPort, intBaud, intData As Integer
Public bytStop As Rs232.DataStopBit
Public bytParity As Rs232.DataParity
Public mComs(3) As Boolean ' ตรวจสอบ Serial Port
```

```
Dim InfoDate As DateTimeFormatInfo
```

```
Dim CardIn_out As Boolean
Dim Sendloop As Integer = 0
Dim TextSend As String = ""
Dim StatusWrite As Boolean
' print document*****
Dim UseFont As New Font("MS Sans Serif", 10)
Dim prDlg As PrintDialog = New PrintDialog
Private prDoc As PrintDocument = New PrintDocument
'*/*****
```

```
Dim Wait As WaitUpdate
Dim Stddate As Boolean
Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Load
    Me.WindowState = FormWindowState.Maximized
    Cursor.Current = Cursors.WaitCursor
```

```
    InfoDate = DateTimeFormatInfo.InvariantInfo
    CardIn_out = False
    InsertMoneyStatus = 0.0
    AdminCheck = False
    StCardIn = False
    Sendloop = 0
    TextSend = ""
    StatusWrite = False
    Wait = New WaitUpdate
    Stddate = True
```

```
    With Conn
        If .State = ConnectionState.Open Then .Close()
        .ConnectionString = strConn
        .Open()
    End With
    Money = 0.0
    TotalMoneyBay = 0.0
    TmrDate.Enabled = True
    ButtonBay.Enabled = False
    FormatDataGrid()
    Me.MainMenu.Enabled = False
    FormatView()
```

```

ClearItemOrder()
Dim i As Integer
For i = 0 To 3
    mComs(i) = IsPortAvailable(i + 1)
Next
InitPort()

AddHandler prDoc.PrintPage, New PrintPageEventHandler(AddressOf
Me.StringToPrint_Print)
prDoc.DocumentName = "ชื่ออาหาร"
prDlg.Document = prDoc
Cursor.Current = Cursors.Default
tmrCardOut.Enabled = True

End Sub
Private Sub StringToPrint_Print(ByVal sender As Object, ByVal e As
PrintPageEventArgs)

    AnyString(e.Graphics, "รหัสนักศึกษา : " + Student_id, 100, 100)
    AnyString(e.Graphics, "ชื่อ : " + NameStudent + " " + SurName, 100, 120)
    AnyString(e.Graphics, "วันที่ : " + DateCurrent.Text, 100, 140)
    AnyString(e.Graphics,
".....", 100, 150)

    AnyString(e.Graphics, "-----"
----", 100, 160)
    AnyString(e.Graphics, "          รายการอาหาร", 100, 175)
    AnyString(e.Graphics, "ราคา", 295, 175)
    AnyString(e.Graphics, "-----"
----", 100, 185)

    Dim i As Integer = 0
    Dim strColumn1 As String = ""
    Dim strColumn2 As String = ""
    Dim CurrentYPosition As Integer = 210
    Dim MainMoneyTol As Double
    For i = 0 To MenuView.Items.Count - 1

        strColumn1 = MenuView.Items(i).SubItems(1).Text
        strColumn2 = MenuView.Items(i).SubItems(2).Text
        AnyString(e.Graphics, " " + strColumn1, 100, CurrentYPosition)
        AnyString(e.Graphics, strColumn2, 300, CurrentYPosition)

        CurrentYPosition = CurrentYPosition + 15
    Next
    CurrentYPosition = CurrentYPosition + 40
    AnyString(e.Graphics, "-----"
---", 100, CurrentYPosition)
    CurrentYPosition = CurrentYPosition + 10
    AnyString(e.Graphics, "          รวมเป็นเงินทั้งหมด", 100, CurrentYPosition)
    AnyString(e.Graphics, TotalBay.Text, 300, CurrentYPosition)
    CurrentYPosition = CurrentYPosition + 10
    AnyString(e.Graphics, "-----"
---", 100, CurrentYPosition)
    CurrentYPosition = CurrentYPosition + 10

```

```

AnyString(e.Graphics, "      ยอดเงินคงเหลือ", 100, CurrentYPosition)
MainMoneyTol = MainMoney.Text
MainMoneyTol = MainMoneyTol - TotalBay.Text
AnyString(e.Graphics, CStr(MainMoneyTol), 300, CurrentYPosition)
CurrentYPosition = CurrentYPosition + 10
AnyString(e.Graphics, "-----
---", 100, CurrentYPosition)

End Sub
Private Sub AnyString(ByVal g As Graphics, ByVal printString As String,
ByVal xPos As Integer, ByVal yPos As Integer)
    Dim anyPoint As New PointF(xPos, yPos)
    g.DrawString(printString, UseFont, Brushes.Black, anyPoint)
End Sub
Private Function InitPort() As Boolean
    intCommPort = 1
    intBaud = 9600
    intData = 8
    bytParity = Rs232.DataParity.Parity_None
    bytStop = Rs232.DataStopBit.StopBit_1
    oCP.Open(intCommPort, intBaud, intData, bytParity, bytStop, 4096)

    tmrRead.Enabled = True

End Function
Private Function IsPortAvailable(ByVal ComPort As Integer) As Boolean
    Try
        oCP.Open(ComPort, 9600, 8, Rs232.DataParity.Parity_None, _
            Rs232.DataStopBit.StopBit_1, 4096)

        oCP.Close()
        Return True
    Catch
        Return False
    End Try
End Function

Private Sub MenuItem6_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem6.Click
    If CheckAdmin() Then
        Me.Close()
    End If
End Sub

Private Sub Button3_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Button3.Click
    If CheckAdmin() Then
        Me.Close()
    End If
End Sub

Private Sub TmrDate_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles TmrDate.Tick
    Dim DateCur As Date
    DateCurrent.Text = ""
    DateCurrent.Text = CStr(DateCur.Today.ToLongDateString) + "      "
```

```

        DateCurrent.Text = DateCurrent.Text +
        CStr(DateCur.Today.Now.ToLongTimeString)

```

```

End Sub

```

```

Private Sub FormatDataGrid()

```

```

    Dim sql As String = ""
    sql = "select * from Foods"
    da = New SqlDataAdapter(sql, Conn)
    ds.Tables.Clear()
    da.Fill(ds, "foods")
    If ds.Tables("foods").Rows.Count = 0 Then
        Me.MainMenu.Enabled = False
    Else

```

```

        Me.MainMenu.Enabled = True
        With MainMenu
            .AllowNavigation = False
            .BackColor = Color.White
            .ReadOnly = True
            .DataSource = ds.Tables("foods")
        End With

```

```

        Dim GrdTs As DataGridViewTableStyle = New DataGridViewTableStyle
        With GrdTs
            .AlternatingBackColor = Color.White
            .BackColor = Color.WhiteSmoke
            .MappingName = "foods"
        End With

```

```

        Dim cs1 As New DataGridViewTextBoxColumn
        With cs1
            .HeaderText = "รหัส"
            .MappingName = "FoodID"
            .ReadOnly = True
            .Width = 50
        End With
        GrdTs.GridColumnStyles.Add(cs1)

```

```

        Dim cs2 As New DataGridViewTextBoxColumn
        With cs2
            .HeaderText = "ชื่ออาหาร"
            .MappingName = "FoodName"
            .ReadOnly = True
            .Width = 150
        End With
        GrdTs.GridColumnStyles.Add(cs2)

```

```

        Dim cs3 As New DataGridViewTextBoxColumn
        With cs3
            .HeaderText = "ราคา"
            .MappingName = "FoodCost"
            .ReadOnly = True
            .Width = 45
        End With

```

```

        GrdTs.GridColumnStyles.Add(cs3)

        With MainMenu
            .TableStyles.Clear()
            .CaptionFont = New Font("MS Sans Serif", 10.0,
FontStyle.Regular)
            .CaptionText = "รายการอาหารทั้งหมด"
            .FlatMode = True
            .TableStyles.Add(GrdTs)
        End With
    End If

End Sub
Private Sub FormatView()
    MenuView.Columns.Add("รหัส", 70, HorizontalAlignment.Left)
    MenuView.Columns.Add("ชื่ออาหาร", 170, HorizontalAlignment.Left)
    MenuView.Columns.Add("ราคา", 55, HorizontalAlignment.Left)

    MenuView.View = View.Details
    MenuView.GridLines = True

End Sub

Private Sub MainMenu_DoubleClick(ByVal sender As Object, ByVal e As
System.EventArgs) Handles MainMenu.DoubleClick
    Dim index As Integer = 0
    Dim i As Integer = 0
    index = MainMenu.CurrentRowIndex
    Dim Temp As String = MainMenu.Item(index, 0)

    Dim sql As String = ""
    sql = "Select * From Foods where FoodID = " + Temp
    da = New SqlDataAdapter(sql, Conn)
    ds.Tables.Clear()
    da.Fill(ds, "Select")
    If ds.Tables("select").Rows.Count <> 0 Then
        Dim Foodid As String
        Dim FoodName As String
        Dim foodcost As String

        Foodid = ds.Tables("select").Rows(0).Item("FoodID")
        FoodName = ds.Tables("select").Rows(0).Item("FoodName")
        foodcost = ds.Tables("select").Rows(0).Item("FoodCost")

        If CheckOverMoney(foodcost) Then

            Dim AnyData() As String
            AnyData = New String() {Foodid, FoodName, foodcost}
            Dim Lview As ListViewItem
            Lview = New ListViewItem(AnyData)
            MenuView.Items.Add(Lview)
            Conculate(foodcost)

        End If
    End If
End Sub

```



```

        If MenuView.Items.Count = 0 Then
            ButtonBay.Enabled = False
        Else
            ButtonBay.Enabled = True
        End If

    End If

End Sub

Private Function CheckOverMoney(ByVal data As String) As Boolean
    Dim tempMon1 As Double = TotalBay.Text
    Dim tempMon2 As Double = MainMoney.Text
    Dim tempMonCur As Double = data
    tempMon1 = tempMon1 + data
    If tempMon1 > tempMon2 Then
        MessageBox.Show("เงินของคุณไม่พอจ่าย กรุณาเพิ่มจำนวนเงิน", "คำเตือน", MessageBoxButtons.OK,
        MessageBoxIcon.Warning)
        Return False
    Else
        Return True
    End If
End Function

Private Function Conculate(ByVal Data As Double) As Boolean
    Dim montemp As Double = Data
    Money = Money + montemp
    TotalBay.Text = CStr(Money)

End Function

Private Sub MenuView_DoubleClick(ByVal sender As Object, ByVal e As
System.EventArgs) Handles MenuView.DoubleClick
    Dim i As Integer
    For i = 0 To MenuView.SelectedItems.Count - 1
        Dim tav As ListViewItem
        tav = MenuView.SelectedItems(i)
        ConculateSub(tav)
        MenuView.Items.Remove(tav)
    Next

    If MenuView.Items.Count = 0 Then
        ButtonBay.Enabled = False
    Else
        ButtonBay.Enabled = True
    End If
End Sub

Private Function ConculateSub(ByVal Data As ListViewItem) As Boolean
    Dim montemp As Double = Data.SubItems(2).Text
    Money = Money - montemp
    TotalBay.Text = CStr(Money)
End Function

Private Sub Button2_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Button2.Click
    ClearItem()
End Sub

Private Sub ClearItem()
    MenuView.Clear()

```

```

Money = 0.0
TotalBay.Text = CStr(Money)
MenuView.Columns.Add("รหัส", 70, HorizontalAlignment.Left)
MenuView.Columns.Add("ชื่ออาหาร", 170, HorizontalAlignment.Left)
MenuView.Columns.Add("ราคา", 55, HorizontalAlignment.Left)

MenuView.View = View.Details
MenuView.GridLines = True
ButtonBay.Enabled = False
End Sub
Private Function IsInstallPrinter() As Boolean
    IsInstallPrinter = False
    If prDoc.PrinterSettings.PrinterName = "<no default printer>" Then
        IsInstallPrinter = False
    Else
        IsInstallPrinter = True
    End If
End Function

Private Sub ButtonBay_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles ButtonBay.Click

    If MessageBox.Show("ต้องการจ่ายค่าอาหารใช่หรือไม่", "คำยืนยัน", MessageBoxButtons.YesNo,
MessageBoxIcon.Question) = DialogResult.Yes Then
        Dim Sql As String = ""
        Dim tempId As String = ""
        Dim tempMon As String = ""
        Dim DateCur As Date
        If IsInstallPrinter() = True Then
            prDoc.Print()
        Else
            MessageBox.Show("กรุณาดัดตั้งเครื่อง Printer ก่อนสั่งพิมพ์เอกสาร", "ผลการตรวจสอบ",
MessageBoxButtons.OK, MessageBoxIcon.Information)
        End If
        Try
            Dim i As Integer
            For i = 0 To MenuView.Items.Count - 1
                tempId = MenuView.Items(i).SubItems(0).Text
                tempMon = MenuView.Items(i).SubItems(2).Text
                Sql = "insert into
BayFoods(Std_ID,Name,SurName,FoodID,Cost,DateCurrent) values ('" + Student_id +
"', '" + NameStudent + "', '" + SurName + "', '" + tempId + "', '" + tempMon + "', '" +
DateCur.Today().ToString("d", InfoDate) + "')"
                da.SelectCommand.CommandText = Sql
                ds.Tables.Clear()
                da.Fill(ds, "insert")

            Next
            Dim tempMon1 As Double = MainMoney.Text
            Dim tempMon2 As Double = TotalBay.Text
            'MainMoney.Text = CStr(tempMon1 - tempMon2)
            Try
                InsertMoneyStatus = tempMon1 - tempMon2
                TextSend = "write" + Chr(13) + Chr(5)
                StatusWrite = True
            End Try
        End Try
    End If
End Sub

```

```

        tmrCardOut.Enabled = False
        tmrWrite.Enabled = True

    Catch ex As Exception

    End Try
    OrderMenu()
    Money = 0.0
    ClearItem()
    Catch ex As Exception
        MsgBox("ข้อมูลผิดพลาด.")
    End Try

End If

End Sub

Private Sub MenuItem2_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem2.Click
    If CheckAdmin() Then
        Dim form As New AddFoods
        If form.ShowDialog() Then
            FormatDataGrid()
        End If
    End If

End Sub

Private Sub MenuItem3_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem3.Click
    If CheckAdmin() Then
        Dim form As New EditFoods
        If form.ShowDialog() Then
            FormatDataGrid()
        End If
    End If

End Sub

Private Sub ClearItemOrder()
    Order.Clear()
    TotalBay.Text = CStr(Money)
    Order.Columns.Add("รหัสนักศึกษา", 80, HorizontalAlignment.Left)
    Order.Columns.Add("ชื่อ", 120, HorizontalAlignment.Left)
    Order.Columns.Add("นามสกุล", 120, HorizontalAlignment.Left)
    Order.Columns.Add("รหัส", 50, HorizontalAlignment.Left)
    Order.Columns.Add("ชื่ออาหาร", 185, HorizontalAlignment.Left)
    Order.Columns.Add("ราคา", 55, HorizontalAlignment.Left)

    Order.View = View.Details
    Order.GridLines = True

End Sub

Private Function OrderMenu() As Boolean

```

```

Dim i As Integer
Dim temp1 As String = ""
Dim temp2 As String = ""
Dim temp3 As String = ""
Dim DateCur As Date
For i = 0 To MenuView.Items.Count - 1
    temp1 = MenuView.Items(i).SubItems(0).Text
    temp2 = MenuView.Items(i).SubItems(1).Text
    temp3 = MenuView.Items(i).SubItems(2).Text

    TotalMoneyBay = TotalMoneyBay + temp3
    TotalMon.Text = TotalMoneyBay

    Dim AnyData() As String
    AnyData = New String() {Student_id, NameStudent, SurName, temp1,
temp2, temp3}
    Dim Lview As ListViewItem
    Lview = New ListViewItem(AnyData)
    Order.Items.Add(Lview)

Next
End Function

Private Sub Order_DoubleClick(ByVal sender As Object, ByVal e As
System.EventArgs) Handles Order.DoubleClick
    If CheckAdmin() Then
        If Student_id = "" Then
            Dim i As Integer
            For i = 0 To Order.SelectedItems.Count - 1
                Dim tav As ListViewItem
                tav = Order.SelectedItems(i)
                'ConculatSub(tav)
                Order.Items.Remove(tav)
            Next
        Else
            Dim i As Integer
            For i = 0 To Order.SelectedItems.Count - 1
                Dim tav As ListViewItem
                tav = Order.SelectedItems(i)
                DeleteMon(tav)
                Order.Items.Remove(tav)
            Next
        End If
    End If
End If

End Sub
Private Function DeleteMon(ByVal tav As ListViewItem) As Boolean
    Dim Mon As String = tav.SubItems(3).Text
    Dim sql As String = ""
End Function
Private Function ChRegisCard() As Boolean
    Dim strlen As Integer = ReadText.TextLength
    Dim loopstr As Integer = 0
    While (loopstr < strlen)
        If ReadText.Text.Chars(loopstr) = Chr(3) Then
            Return True
        End If
    End If
End Function

```

```

        loopstr = loopstr + 1
    End While
    Return False
End Function
Private Function CheckCardIn() As Boolean
    If ChRegisCard() Then

        Dim CardIn As String = ReadText.Text
        SmartHealth = ""
        SmartStudent = ""
        SmartMoney = ""

        CardIn = CardIn + Chr(4)
        Dim LoopIn As Integer = 0
        While CardIn.Chars(LoopIn) <> Chr(4)
            If CardIn.Chars(LoopIn) = Chr(2) Then
                LoopIn = LoopIn + 1
                While CardIn.Chars(LoopIn) <> Chr(3)
                    SmartStudent = SmartStudent + CardIn.Chars(LoopIn)
                    LoopIn = LoopIn + 1
                End While
                LoopIn = LoopIn + 1
                While CardIn.Chars(LoopIn) <> Chr(3)
                    SmartHealth = SmartHealth + CardIn.Chars(LoopIn)
                    LoopIn = LoopIn + 1
                End While
                LoopIn = LoopIn + 1
                While CardIn.Chars(LoopIn) <> Chr(3)
                    SmartMoney = SmartMoney + CardIn.Chars(LoopIn)
                    LoopIn = LoopIn + 1
                End While
                LoopIn = LoopIn + 1
            End If
            LoopIn = LoopIn + 1
        End While

        CheckPassword()
    Else
        MsgBox("บัตร ยังไม่ป้อน")
    End If

    Return True
End Function
Private Function CheckPassword() As Boolean
    Dim lengthdata As Integer = SmartStudent.Length
    Dim loopdata As Integer = 0

    Dim datapwd As String = ""
    Dim dataIdstd As String = ""
    Dim tempname As String = ""
    Dim tempsurname As String = ""
    'MsgBox(SmartStudent)

    While (loopdata < lengthdata)
        While (SmartStudent.Chars(loopdata) <> Chr(64))
            datapwd = datapwd + SmartStudent.Chars(loopdata)
            loopdata = loopdata + 1
        End While
    End While

```

```

    End While
    loopdata = loopdata + 1

    While (SmartStudent.Chars(loopdata) <> Chr(64))
        dataIdstd = dataIdstd + SmartStudent.Chars(loopdata)
        loopdata = loopdata + 1
    End While
    loopdata = loopdata + 1

    While (SmartStudent.Chars(loopdata) <> Chr(64))
        tempname = tempname + SmartStudent.Chars(loopdata)
        loopdata = loopdata + 1
    End While
    loopdata = loopdata + 1

    While (loopdata < lengthdata)
        tempsurname = tempsurname + SmartStudent.Chars(loopdata)
        loopdata = loopdata + 1
    End While
    loopdata = loopdata + 1
End While

Student_id = dataIdstd
NameStudent = tempname
SurName = tempsurname
Dim temp As String = DecodePwd(datapwd)
CheckPassword(temp)

End Function

Private Function DecodePwd(ByVal CodeData As String) As String

    Dim i As Integer = 0
    Dim tenp0 As Integer = Asc(CodeData.Chars(0))
    Dim tenp1 As Integer = Asc(CodeData.Chars(1))
    Dim tenp2 As Integer = Asc(CodeData.Chars(2))
    Dim tenp3 As Integer = Asc(CodeData.Chars(3))
    Dim CodeData1 As String = ""
    i = 4

    CodeData1 = Chr(tenp0 + 10)
    CodeData1 = CodeData1 + Chr(tenp1 + 15)
    CodeData1 = CodeData1 + Chr(tenp2 + 10)
    CodeData1 = CodeData1 + Chr(tenp3 + 17)
    While i < CodeData.Length
        CodeData1 = CodeData1 + CodeData.Chars(i)
        i = i + 1
    End While
    Return CodeData1

End Function

Public Function CheckPassword(ByVal Data As String) As Boolean
    DataOfCard = Data
    Dim Forms As New LOGIN
    If Forms.ShowDialog() Then
        If Pwd Then

```

```

        If CInt(SmartMoney) < 5 Then
            StCardIn = True
            MessageBox.Show("บัตรของคุณมีเงินไม่พอจ่าย กรุณาเติมเงิน", "ถ้าเตือน",
MessageBoxButtons.OK, MessageBoxIcon.Warning)
            MainMoney.Text = 0
            tmrCardOut.Enabled = True
        Else
            Me.MainMenu.Enabled = True
            tmrCardOut.Enabled = True
            StCardIn = True
            MainMoney.Text = SmartMoney
        End If

    End If
End If
'MsgBox(DataOfCard)
End Function

Private Sub tmrRead_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles tmrRead.Tick
    Try
        While (oCP.Read(1) <> -1)
            Me.tmrRead.Enabled = False
            If (Chr(oCP.InputStream(0)) <> Chr(4)) Then
                ReadText.Text = ReadText.Text + Chr(oCP.InputStream(0))
            End If
            CardIn_out = True
        End While

        Catch exc As Exception
            If StatusWrite Then
                StatusWrite = False
                TextSend = SmartStudent + Chr(3) + SmartHealth + Chr(3) +
CStr(InsertMoneyStatus) + Chr(3) + Chr(4) + Chr(13) + Chr(5)
                Me.tmrWrite.Enabled = True
                Sendloop = 0
            Else
                If CardIn_out And ReadText.Text.Length > 100 Then
                    CheckCardIn()
                    CardIn_out = False
                Else
                    Me.tmrRead.Enabled = True
                End If
            End If

        End Try
    End Sub

Private Sub MenuItem5_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem5.Click

    If CheckAdmin() Then
        Dim Forms As New RegisMoney
        If Forms.ShowDialog() Then
            If InsertMoneyStatus Then
                Me.MainMenu.Enabled = True
                TextSend = "write" + Chr(13) + Chr(5)
            End If
        End If
    End If
End Sub

```

```

        'MsgBox(MainMoney.Text)
        StatusWrite = True
        Wait.Show()
        tmrCardOut.Enabled = False
        tmrWrite.Enabled = True
    Else
        MsgBox("ไม่สามารถเติมเงินได้")
    End If
End If
End If

End Sub

Private Sub tmrCardOut_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles tmrCardOut.Tick

    Dim out As Boolean = False
    Try
        While (oCP.Read(1) <> -1)
            Me.tmrCardOut.Enabled = False
            ReadText.Text = ReadText.Text + Chr(oCP.InputStream(0))
            out = True
        End While

    Catch exc As Exception
        If out Then
            If ReadText.Text.Length > 15 Then
                If ReadText.Text.Chars(1) = Chr(82) Then
                    SmartStudent = ""
                    SmartHealth = ""
                    SmartMoney = ""
                    out = False
                    CardIn_out = False
                    Me.MainMenu.Enabled = False
                    StCardIn = False
                    MainMoney.Text = 0
                    'tmrWrite.Enabled = False
                    Me.tmrCardOut.Enabled = False
                    Me.tmrCardOut.Stop()
                    'MsgBox("Remove Card out")
                    ClearItem()
                    ReadText.Text = ""
                    tmrRead.Enabled = True
                Else
                    out = False
                    Me.tmrCardOut.Enabled = True
                End If
            Else
                out = False
                Me.tmrCardOut.Enabled = True
            End If
        Else
            Me.tmrCardOut.Enabled = True
            ReadText.Text = Chr(13)
        Try

```



```

        oCP.Write(Encoding.ASCII.GetBytes(Me.ReadText.Text &
Chr(13)))
        Catch ex As Exception
            MessageBox.Show("Unable to write to comm port")
            'Me.tmrWrite.Enabled = False
        Finally
            'ReadText.Text = ""
        End Try
    End If
End Try

End Sub
Private Function CheckAdmin() As Boolean
    Dim Form As New Admin
    If Form.ShowDialog Then
        If AdminCheck Then
            Return True
        Else
            Return False
        End If
    End If
End Function

Private Sub tmrWrite_Tick(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles tmrWrite.Tick
    If TextSend.Chars(Sendloop) <> Chr(5) Then
        ReadText.Text = TextSend.Chars(Sendloop)
        Sendloop = Sendloop + 1

        Try
            oCP.Write(Encoding.ASCII.GetBytes(Me.ReadText.Text))
            'WriteMessage(ReadText.Text, True)
        Catch ex As Exception
            MessageBox.Show("Unable to write to comm port")
            Me.tmrWrite.Enabled = False
        Finally
            ReadText.Text = ""
        End Try
    Else
        Me.tmrWrite.Enabled = False
        Sendloop = 0
        If StatusWrite Then
            Me.tmrRead.Enabled = True
        Else
            'MsgBox("Register Successfull.....")
            Wait.Close()
            Wait = New WaitUpdate
            MainMoney.Text = InsertMoneyStatus
            Me.tmrCardOut.Enabled = True
        End If
    End If
End Sub

Private Sub MenuItem4_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem4.Click
    If CheckAdmin() Then

```

```
        Dim form As New EditFoods
        If form.ShowDialog() Then
            FormatDataGrid()
        End If
    End If
End Sub

Private Sub MenuItem8_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MenuItem8.Click
    Dim Form As New Search
    Form.Show()
End Sub

End Class
```

ภาคผนวก ข. มาตรฐานชิพการ์ดและสมาร์ทการ์ด

● **The ISO 7816 Smart Card Standard**

ISO7816 document	Description
ISO 7816-1	Physical characteristics
ISO 7816-2	Dimensions and location of the contacts
ISO 7816-3	Electronic signals and transmission protocols
ISO 7816-4	Industry commands for interchange
ISO 7816-5	Number system and registration procedure for application identifiers
ISO 7816-6	Interindustry data elements

● **ISO 7816-1 Physical characteristics**

The characteristics of a card when it is bent or flexed. This is to make sure that plastic cards with embedded chips are manufactured in a way that guarantees flawless operation over the expected life time of a card. Connections between the surface connectors and the I/O pins of the embedded silicon die must be maintained and withstand mechanical stress. Bending and flexing procedures are standardised in ISO 7816

● **ISO 7816-2 Dimensions and location of the contacts**

The integrated circuit card (ICC) has 8 electrical contacts. They are referred to as C1 through C8. However, not all 8 contacts are electrically connected to the embedded microprocessor chip and therefore unused at the present time.

The following table contains the contact definition according to ISO 7816-2

Contact	Designation	Use
C1	Vcc	Power connection through which operating power is supplied to the microprocessor chip in the card
C2	RST	Reset line through which the IFD can signal to the smart card's microprocessor chip to initiate its reset sequence of instructions
C3	CLK	Clock signal line through which a clock signal can be provided to the microprocessor chip. This line controls the operation speed and provides a common framework for data communication between the IFD and the ICC
C4	RFU	Reserved for future use
C5	GND	Ground line providing common electrical ground between the IFD and the ICC
C6	Vpp	Programming power connection used to program EEPROM of first generation ICCs.
C7	I/O	Input/output line that provides a half-duplex communication channel between the reader and the smart card
C8	RFU	Reserved for future use

● **ISO 7816-3 Electronic signals and transmission protocols Electrical Signals Description**

I/O : Input or Output for serial data to the integrated circuit inside the card.

VPP : Programming voltage input (optional use by the card).

GND : Ground (reference voltage).

CLK : Clocking or timing signal (optional use by the card).

RST : Either used itself (reset signal supplied from the interface device) or in combination with an internal reset control circuit (optional use by the card). If internal reset is implemented, the voltage supply on Vcc is mandatory

VCC : Power supply input (optional use by the card). NOTE - The use of two remaining contacts will be defined in the appropriate application standards

Answer to Reset

Two types of transmissions are considered:

1. Asynchronous transmission:

In this type of transmission, characters are transmitted on the I/O line in an asynchronous half duplex mode. Each character includes an 8bit byte.

2. Synchronous transmission:

In this type of transmission, a series of bits is transmitted on the I/O line in half duplex mode in synchronisation with the clock signal on CLK.

- **ISO 7816-4 Industry commands for interchange**

Unquestionably the most widely talked-about smart card API is that set forth in the ISO 7816-4 standard, which is discussed at length in Chapter 4, “Smart Card Commands.” There are 18 basic interindustry commands on this API:

ReadBinary(byte *fileId*, short *offset*, byte *buffer*[])

WriteBinary(byte *fileId*, short *offset*, byte *buffer*[])

UpdateBinary(byte *fileId*, short *offset*, byte *buffer*[])

EraseBinary(byte *fileId*, byte *offset*)

ReadRecord(byte *record_number*, byte *mode*, byte *buffer*[])

WriteRecord(byte *record_number*, byte *mode*, byte *buffer*[])

AppendRecord(byte *record_number*, byte *mode*, byte *buffer*[])

UpdateRecord(byte *record_number*, byte *mode*, byte *buffer*[])

GetData(short *mode*, byte *buffer*[])

PutData(short *mode*, byte *buffer*[])

SelectFile(byte *mode*, byte *info*, byte *name*[])

Verify(byte *mode*, byte *key*[])

InternalAuthenticate(byte *algorithm*, byte *mode*, byte *challenge*[])

ExternalAuthenticate(byte *algorithm*, byte *mode*, byte *response*[])

GetChallenge(byte *challenge*[])

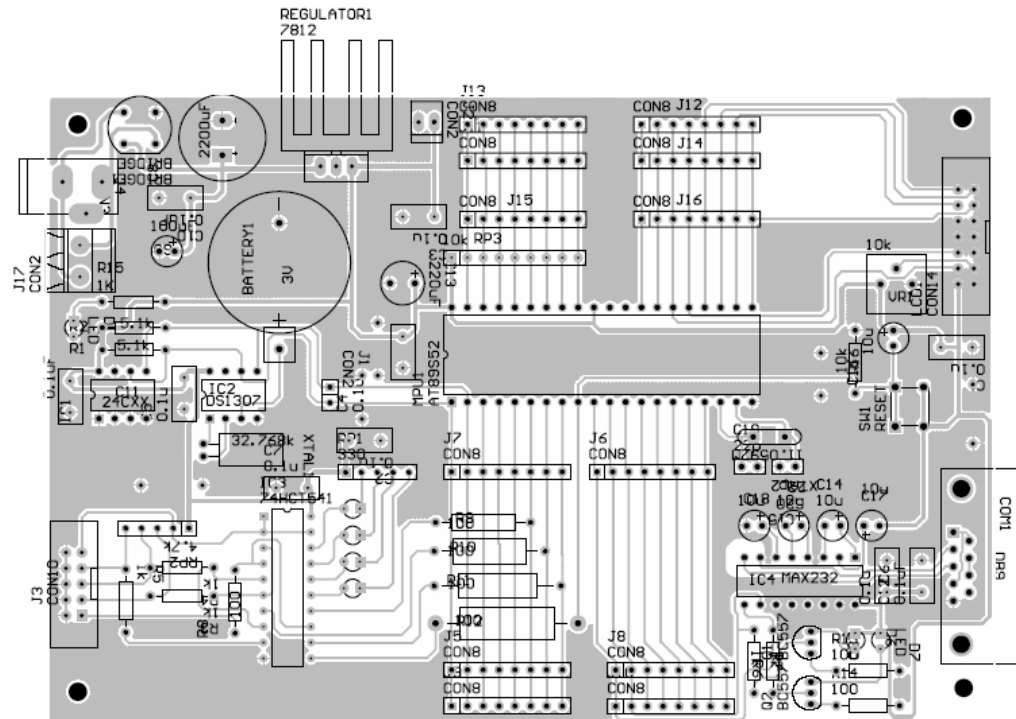
ManageChannel(byte *operation*, byte *channel_number*)

GetResponse(byte *response*[]);

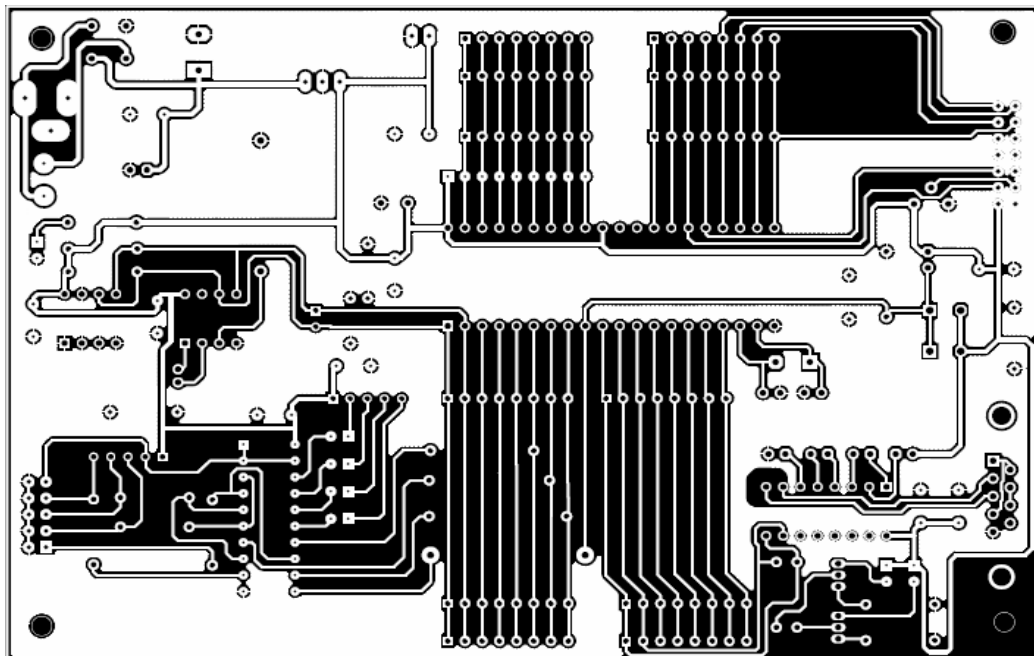
Envelope(byte *buffer*[])

As far as we know, there is no smart card on the market—nor has there ever been—that implements all the ISO 7816-4 commands with all the generality and capability written into the standard. Nevertheless, the ISO 7816-4 commands have served as a reference model for smart card interfacing and have to at least some degree enabled card interoperability. Many off-the-shelf smart cards implement some variant of most of these commands.

ภาคผนวก ค. วงจรลายปริ้นท์ และวัสดุที่ใช้



การลงอุปกรณ์



ลายวงจร

● รายการอุปกรณ์

ชื่ออุปกรณ์	จำนวน
IC	
AT89S52	1 ตัว
MAX232	1 ตัว
7805	1 ตัว
X'TAL	
11.059MHz	1 ตัว
Resistor	
100 Ohm	2 ตัว
1 KOhm	2 ตัว
10 KOhm	1 ตัว
Resistor Network	
10K 9PIN	1 ตัว
Transistor	
BC557	2 ตัว
LED	2 ตัว
DIODE BRIDGE	
200V 1.5A	1 ตัว
Capacistor	
2200 uF/35V	1 ตัว
100 uF/35V	1 ตัว

220uF/16V	1 ตัว
10uF/16V	5 ตัว
0.1uF/50V Multilayer	5 ตัว
22pF/50V	2 ตัว
DB9/F PCB	1 ตัว
RESET SW	1 ตัว